DRAFT — Ctrl_+_Alt_+_Defeat modeling memo

Suffolk University · ISOM 835 Fall 2026 · Head of Revenue Management
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1. What You Tried

Ten five-fold experiments (seed 835) compared logistic regression, decision trees, random forests and gradient boosting. Baseline AUC was 0.88936; corrected categorical IDs reached 0.90740 and engineered logistic 0.91597. Exploratory boosting reached 0.94471. Conservative regularized boosting reached 0.88323, with fold SD 0.00342. These are validation results, not Kaggle scores.

2. What Worked

The largest controlled same-model feature improvement excluding uncertain-field inclusion was categorical agent/company handling: +0.01804 AUC. Treating IDs as labels avoids imposing numeric order. The largest comparable model-family change on conservative engineered inputs was logistic to random forest: +0.06721; it changes model capacity as well. Neither comparison establishes production benefit without timing review.

3. What Leaked or Almost Did

Known outcomes reservation_status, reservation_status_date and assigned_room_type were verified absent. The conservative candidate drops booking_changes, days_in_waiting_list, total_of_special_requests, agent, company, country and deposit_type. Adding them to fixed-config boosting raised AUC 0.07131; that is not proof of leakage. Booking creation is provisional; remaining fields such as ADR/parking still need timestamp evidence. No model is certified leak-free.

4. Your Score

Public Kaggle AUC: PENDING confirmed submission. Private Kaggle AUC: PENDING official reveal. Best conservative CV AUC: 0.88323; exploratory CV: 0.94471. No leaderboard comparison is possible. Candidate OOF Brier score is 0.12942; ranking and calibration answer different questions.

5. With Two More Weeks

Test booking-time validation and calibrate probabilities using training-only nested folds. Expected AUC improvement: not yet estimated; changes could be negative under time shift. Timing evidence is a prerequisite. Reliable probabilities could support expected cancellation counts, with capacity/cost review before decisions.

AI assistance: ChatGPT and OpenAI coding tools supported coding, analysis and drafting. Team review is pending. DRAFT; not ready for Canvas upload.